Kinematic Modelling and MATLAB-Based Analysis of Delta Robot Algorithm

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1 Introduction

Robot manipulators are widely used across industrial, medical and educational domains because of their precision, repeatability and programmability [8]. Common applications include pick-and-place operations, assembly tasks, welding, painting, and laboratory automation. In educational environments, robotic platforms are used to teach fundamentals of kinematics, motion planning, robotics programming, control, and integration [1].

The Delta Robot Kit is a desktop serial manipulator consisting of four revolute joints (21-24), four actuated servomotors (motors 1-4 on schematic) and 3D printed in both joint and Cartesian coordinate systems, enabling flexible motion control and trajectory planning. The robot supports multiple motion modes [9] such as point-to-point (PTP), linear (MOVL), and arc-based trajectories, making it suitable for both basic and advanced robotics experiments.

Unlike a classical serial robot, the Delta Robot Mechanism Kit can be categorized as a serial articulated robot arm. Compared to parallel manipulators, serial manipulators offer a larger workspace and greater flexibility, but may suffer from lower stiffness and accumulated positioning errors. Compared to SCARA robots, the Delta robot provides greater vertical flexibility but may have reduced speed and precision in planar tasks.

The Delta Robot Kit is particularly suitable for educational use because it integrates kinematics, control, and trajectory planning into a compact platform. Its limitations include reduced payload capacity (250 g), limited reach (340 mm), and lower industrial-grade repeatability compared to high-end robotic systems [3]. The aim of this project was to develop an analytical and MATLAB-based model of the Delta Robot Kit, derive its forward and inverse kinematics, inverse kinematics, Jacobian, singularities and calibration behaviour.

List of Symbols

- θ_i – base rotation angle
- θ_2 – shoulder joint angle
- θ_3 – elbow joint angle
- θ_4 – end-effector rotation angle
- L_1, L_2, L_3 – main arm link lengths

2 Kinematic Modelling

2.1 Robot configuration and assumptions

The kinematic model therefore treats the robot as a 6R serial manipulator, defined by the revolute rotation, θ_i and θ_2 and θ_3 are the shoulder and elbow controlling tool orientation.

Although the physical robot includes a parallelogram mechanism [4] that helps maintain end-effector orientation, the kinematic model used in this work represents that behaviour through an equivalent serial kinematic representation [6]. This passive element is not treated as an additional actuated degree of freedom, but only as a mechanism that keeps the end-effector the geometric constraint.

Figure 1: Appearance of the Delta Robot Kit showing the physical arrangement of the four revolute joints used in the kinematic model. Adapted from [16]

2.2 Denavit–Hartenberg parameterization

Figure 2: Joint coordinate system of the Delta Robot Kit. Frame 0 is the base, revolute joints considered in the kinematic model. Adapted from [16].

Figure 3: Denavit–Hartenberg frame assignment used for the kinematic modelling of the Delta Robot Kit. Adapted from [16]

The Denavit–Hartenberg parameters table derived from the frame assignment is shown in Table 1.

Table 1: Denavit–Hartenberg parameter table derived from the frame assignment.

Link	a_i (mm)	α_i	d_i (mm)	θ_i
1 – Base Arm	0	$90°$	150	θ_1
2 – Rear Arm	0	$0°$	0	θ_2
3 – Rear Arm	150	$0°$	0	θ_3
4 – Effector/Platform link	0	$0°$	0	$-\theta_2 - \theta_3$
5 – End-Effector	0	$0°$	0	θ_4

The coordinate frames were assigned according to the standard Denavit-Hartenberg convention [6]. Each z-axis was aligned with the axis of rotation of the corresponding joint. Shifts each x-axis was defined along the common normal between adjacent joint axes. The physical parallelogram mechanism ensures that the end-effector remains parallel to the base platform. This passive mechanism maintains the end-effector orientation approximately constant. This passive item is not independent joint and does not increase the number of actuated degrees of freedom [5].

The Denavit Hartenberg convention provides a systematic framework for assigning frames and deriving homogeneous transformations for serial manipulators [2, 3].

2.3 Forward kinematics

The forward kinematics of the Delta Robot Mechanism Kit are derived using the Denavit Hartenberg convention. A homogeneous transformation matrix was assigned to each joint in order to describe the position and orientation of the link frame relative to the preceding frame. The overall transformation from the base to the end-effector was then obtained by multiplying the individual link transformations [6, 7].

For the equivalent kinematic model used in this project, the total transformation is written as

$$T_{04} = A_1 A_2 A_3 A_4$$

where A_4 denotes the passive transformation introduced to represent the parallelogram constraint in the physical robot. This passive term is included only to preserve the geometric behaviour of the end-effector and does not correspond to an additional actuated degree of freedom.

The forward kinematics transformation matrix provides both the orientation of the end-effector and its Cartesian position relative to the base. The position vector is extracted from the fourth column of the resulting matrix and used as the reference coordinate for subsequent inverse kinematics, Jacobian and singularity analysis.

TODO: This is wrong as we need L1 which is non-zero

$$T_{03} = \begin{bmatrix} \cos\theta_1 \cdot S_2 & 0 & -\sin\theta_1 \cdot S_2 & \cos\theta_1 [L_2 \cos\theta_2 + L_3 \cos(\theta_2 + \theta_3)] \\ \sin\theta_1 \cdot S_2 & 0 & \cos\theta_1 \cdot S_2 & \sin\theta_1 [L_2 \cos\theta_2 + L_3 \cos(\theta_2 + \theta_3)] \\ 0 & 1 & 0 & h + L_2 \sin\theta_2 + L_3 \sin(\theta_2 + \theta_3) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where: θ_1 is the base rotation, θ_2 is the shoulder joint angle, θ_3 is the elbow joint angle, θ_4 is the tool rotation, L_2 is the rear arm length, L_3 is the forearm length, h is the base height. Note that L_1 is not included in the kinematics because it is a constant term that does not affect the velocity mapping.

Analytical Jacobian

The analytical Jacobian matrices of $J_v(q)$ and is given by

$$J_v(q) = \begin{bmatrix} -\sin\theta_1 \cdot [\cos\theta_2 [L_2 + L_3 \sin(\theta_3)] - L_3 \sin\theta_2 \cdot \cos(\theta_3)] & -L_3 \sin(\theta_3 + \theta_2) \cos\theta_1 \\ \cos\theta_1 \cdot [\cos\theta_2 [L_2 + L_3 \sin(\theta_3)] - L_3 \sin\theta_2 \cdot \cos(\theta_3)] & -L_3 \sin(\theta_3 + \theta_2) \sin\theta_1 \\ 0 \quad L_3 \cos(\theta_3) + L_2 \cos(\theta_2 + \theta_3) & L_3 \cos(\theta_2 + \theta_3) \end{bmatrix}$$

The base offset h appears in the forward kinematics of z, but it does not appear in the Jacobian because it is a constant term. Its derivative with respect to all joint variables is zero.

Singularities

A singularity occurs when the determinant of the analytical Jacobian becomes zero:

$$\det(J) = L_3 L_2 \sin(\theta_3) r = 0$$

This expression shows that singularities occur when either $\sin(\theta_3) = 0$ or $r = 0$. The condition $\sin(\theta_3) = 0$ corresponds to the arm being fully stretched or fully folded, so that the links become collinear. The condition $r = 0$ corresponds to the wrist lying on the base axis, where the manipulator loses motion authority in certain horizontal directions. Near these configurations, the velocity mapping becomes ill-conditioned and large joint velocities may be required to generate small end-effector motions [5].

2.5 Inverse kinematics

The inverse kinematics problem was solved using a geometric approach, because it provides closed-form solutions and is well suited to two-link planar arm structures [2, 3].

Figure 4: Top-view geometry used to compute the base joint angle θ_1.

The base joint angle was obtained from the projected position in the x–y plane:

$$\theta_1 = \operatorname{atan2}(y, x)$$

The radial distance in the horizontal plane is then $r = \sqrt{x^2 + y^2}$. The remaining problem is reduced to a planar two-link geometry in the $r - z$ plane [2]. A geometric inverse-kinematics method was selected because it provides closed-form solutions and is well suited to two-link planar arm structures [2, 3].

TODO: Start as to why atan2 plus a link to the matlab docs.

Figure 5: Planar geometric construction used for the inverse kinematics derivation.

Figure 5 shows the planar geometry used to derive the shoulder and elbow angles. The coordinates are defined for reachable targets, where the shoulder, elbow, and end-effector form a triangle. Accordingly, the shoulder angle is given by

$$\theta_2 = \alpha + \beta$$

for one branch and

$$\theta_2 = \alpha - \beta$$

for the other. The elbow angle θ_3 is then obtained from the corresponding trigonometric relation, giving two valid inverse-kinematics solutions for the same target position.

3 Materials and Methods

3.1 Kinematic Modelling Approach

The kinematic model of the Delta Robot Mechanism Kit was developed using the Denavit–Hartenberg (DH) convention, which provides a systematic framework for describing the spatial relationship between adjacent links. Each joint was modelled as a revolute joint with one rotation about its corresponding z-axis. Passive mechanical elements, such as the parallelogram mechanism, were not treated as additional actuated degrees of freedom. Instead, they were accounted for through the geometric constraints they impose on the end-effector.

This approach was preferred over numerical methods due to its computational simplicity and suitability for real-time applications.

3.3 Dynamic modelling approach

The dynamic behaviour of the manipulator was considered using the standard rigid-body robot dynamics formulation [6, 7]:

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q)$$

where τ is the vector of joint torques, $M(q)$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ represents Coriolis and centrifugal effects, and $G(q)$ represents gravity torque.

In the present project, the dynamic equations were incorporated at a conceptual level modelling level and were not used for simulation or control in MATLAB. However, a complete high-fidelity dynamic implementation was limited by the lack of reliable mass, inertia, and friction parameters for the physical robot. Future work remains from the current kinematic and control implementation, focused on establishing the dynamic form of the equations and explaining how such terms would be integrated into a MATLAB model for future extension.

3.4 MATLAB Implementation

The robot model was implemented in MATLAB using a modular approach. Individual functions were created for the Denavit-Hartenberg transformations, forward kinematics evaluation, inverse kinematics solution, Jacobian computation, and workspace visualisation. Symbolic computation was used where appropriate to derive compact expressions for the forward kinematics and Jacobian, while numerical functions were used for plotting and simulation.

The implementation was designed so that a simple Cartesian motion could be provided as input, the corresponding joint angles could be computed through inverse kinematics, and the resulting configuration could be visualised through forward kinematics reconstruction. Additional checks were included to detect unreachable targets and singular configurations.

3.5 Testing Framework

A structured testing framework was used to validate the correctness of the mathematical model and its MATLAB implementation. Forward kinematics was tested by predefining known joint configurations and comparing the resulting end-effector positions with hand-calculated values. Inverse kinematics was validated by solving for joint variables from a target Cartesian position and checking whether the forward kinematics reconstructed the original position. Reachable and unreachable targets were used to verify singularity detection and rejection for unreachable targets. Together, these checks ensured internal consistency between the analytical derivation and the computational implementation.

4 Results and Validation

The MATLAB model was validated using forward-kinematics, inverse-kinematics and singularity test cases. Forward kinematics was checked by comparing computed end-effector positions with hand-calculated reference values. Inverse kinematics was verified by reconstructing joint configurations from target positions. Additional test cases were used to confirm singularity detection and rejection of unreachable targets.

Table 2: Validation of forward and inverse kinematics test results table

Test case	Input	Output	Expected
Forward kinematics	$\theta_1 = 0.785$ rad, $\theta_2 = 0.785$ rad, $\theta_3 = 0.785$ rad	$(x, y, z) = (0.0487, 0.106, 0.0487)$	$(x, y, z) = (0.0487, 0.106, 0.0487)$
Inverse kinematics	Target $(0.1837, 0.1061, 0.2591)$ m	$\theta_1 = 0.5236$, $\theta_2 = 0.7854$, $\theta_3 = 1.0472$	$\theta_1 = 0.5236$, $\theta_2 = 0.7854$, $\theta_3 = 1.0472$
Singularity detection	$\sin(\theta_3) = 0$	Singularity detected	Singularity detected
Unreachable target	$(0.5, 0.5, 0.5)$ m	Unreachable	Unreachable

The validation results show good agreement between the analytical model and the MATLAB implementation. In particular, the consistency between the forward-kinematics output and the inverse kinematics reconstruction shows that the position mapping is internally correct. The singularity and unreachable-target tests further confirm that the implementation correctly handles important edge cases within the robot workspace.

Figure 6: Successful inverse kinematics solution using the MATLAB-based UI.

Figure 7: Detection of unreachable target in inverse kinematics.

The system correctly identifies when a target position lies outside the robot's reachable workspace. In this example, the computed distance d exceeds the maximum reach of the arm ($d = 1.0$ m, $L_{max} = 0.36$ m), indicating a non-reachable target. This demonstrates that the inverse kinematics implementation includes proper workspace validation and prevents physically invalid solutions.

5 Discussion and Conclusion

This project developed and validated a kinematic model of the Delta Robot Mechanism Kit using the Denavit–Hartenberg convention and implemented the analysis in MATLAB. Forward kinematics, inverse kinematics, Jacobian computation, and workspace visualisation were developed and validated, and singularities were identified from the analytical model. The resulting MATLAB implementation was used to reproduce valid robot positions and support target-based inverse kinematics.

Overall, the main contribution was representing the physical parallelogram mechanism of the robot within a tractable analytical framework. This was achieved by treating the passive mechanism as an equivalent serial kinematic model, which preserved the geometric behaviour of the end-effector while avoiding a more complex constraint without introducing an additional actuated degree of freedom. Analytical Jacobian was derived and its inverse formulation in a form suitable for reliable computation; this was solved by reducing the problem to a planar geometric construction in the $r - z$ plane. A further limitation concerned dynamics, since accurate link masses, centres of mass and inertia tensors were not available for the physical robot, so dynamic modelling remained conceptual.

Despite these limitations, the final model provides a robust analytical and computational framework for studying the motion of the Delta Robot Mechanism Kit. The implementation makes the model suitable for forward kinematics, inverse kinematics, singularity detection and validation testing in MATLAB. Future work could extend the model to include dynamic parameters, actuator limits, dynamic parameters, actuator constraints, joint limits and closed-loop control in order to improve realism and practical applicability.

6 References

References

[1] P. Corke, *Robotics, Vision and Control: Fundamental Algorithms in MATLAB*, Springer, 2nd edition, 2017.

[2] J. J. Craig, *Introduction to Robotics: Mechanics and Control*. Pearson Education, Upper Saddle River, 3rd edition, 2005.

[3] J. Denavit and R. S. Hartenberg, A kinematic notation for lower-pair mechanisms based on matrices. *Journal of Applied Mechanics*, 22:215–221, 1955.

[4] Elecfreaks. Delta robot kit: Code (Dobot/ot-based), n.d. [Online]. Available: https://elecfreaks.com/product-manual/Delta%20Robot%20Kit%20Code%20Dobot%20ot-based.pdf.

[5] Elecfreaks. Delta robot kit: Assembly tutorial, n.d. [Online]. Available: https://elecfreaks.com/product-manual/Delta%20Robot%20Kit%20Assembly%20Tutorial.pdf.

[6] Elecfreaks. Delta robot kit: Kinematics and inverse kinematics for robot arm, n.d. [Online]. Available: https://ak.autodesk.com/adp/public/attachments/pt-BR/view/inline/5ed7d7ed-0ed1-4ed1-9ed1-0ed1ed1ed1ed.pdf.

[7] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo. *Robotics: Modelling, Planning and Control*. Springer London, 2009.

[8] M. W. Spong, S. Hutchinson, and M. Vidyasagar. *Robot Modeling and Control*. Wiley, Hoboken, 2006.